

Nathan David Obeng-Amoako, E.I.T.

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RESEARCH INTERESTS

- Transit Signal Priority
- Traffic Simulations / Digital Twins
- Intelligent Transportation Systems
- Multimodal Corridor Planning
- Bicycle Network Analysis
- Geospatial Methods in Transportation

EDUCATION

Doctor of Philosophy in Civil Engineering (Transportation) Sep 2023 – Present

Northeastern University, Boston, MA (GPA: 4.00/4.00)

Research Advisor: Professor Peter Furth

Master of Science in Civil Engineering (Transportation) Sep 2023 – Apr 2025

Northeastern University, Boston, MA (GPA: 4.00/4.00)

Bachelor of Science in Civil Engineering Aug 2018 – Sep 2022

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

Recognition: Best Graduating Student, Class of 2022

PROFESSIONAL EXPERIENCE

Transportation Engineering Intern, Fehr & Peers, Oakland, CA Apr 2026 – Jun 2026

- Supported transit signal priority and complete streets projects for state/municipal clients
- Contributed to metro station area studies and transit-oriented development analysis in the East Bay
- Developed federal grant materials for the City of Palo Alto under the OBAG 4 funding cycle
- Conducted GIS and active transportation analysis for corridor and safety planning projects

Program Assistant, Sustainable Transportation Summer Academy, Delft, Holland Jun 2024 – Aug 2026

- Provided logistical, academic, and evaluative support to students over three summers
- Assisted in teaching Vision Zero safety principles to 25 Northeastern University students
- Explored car-free and car-lite zone strategies to enhance active/public transportation

Assistant Transportation Engineer, BHM Construction International UK Ltd. Oct 2022 – Jul 2023

- Conducted over 20 quality control tests daily on construction materials to ensure strict standard compliance
- Supported site reporting and coordination with a team of 5 engineers to improve workflow efficiency

Civil Engineering Trainee, China Henan International Corporation Ltd. Sep 2021 – Dec 2021

- Supervised 15+ artisans daily to ensure strict adherence to project specifications and timelines
- Contributed to precise structural detailing for reinforced concrete components

SPONSORED RESEARCH PROJECTS

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- Project:** Digital Twin for Adaptive Traffic Signal Control **Aug 2026 – Present**
Funding Agency: StreetIQ, LLC (an AI-powered infrastructure intelligence platform)
- Building and calibrating a Vissim microsimulation of Boston arterial corridors as a testbed for adaptive signal control algorithm development
 - Developing external signal control software via Vissim's API to enable iterative, closed-loop algorithm testing and validation against existing control
- Project:** Bicycle Level of Traffic Stress at Intersections (xLTS) **Jun 2024 – Aug 2025**
Funding Agency: United States Department of Transportation (USDOT)
- Assessing the impact of high-stress crossings on bike network connectivity
 - Developing algorithms and data structures for traffic stress network analysis
- Project:** Accessible Bus Stop Design in the Presence of Bike Lanes **Sep 2023 – Jun 2024**
Funding Agency: Massachusetts Department of Transportation (MassDOT)
- Studied bus rider and bicyclist behavior at floating and constrained bus stops
 - Proposed design improvements to reduce bus-rider-and-bicyclist conflicts

PEER-REVIEWED JOURNAL PUBLICATIONS

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- 2025 1. Peter Furth, Ray Saeidi-Razavi, **Nathan David Obeng-Amoako** and Milad Tahmasebi. TSP-Friendly Underlying Traffic Signal Control: An Essential Complement of Transit Signal Priority. *Future Transportation* (2025), doi: <https://doi.org/10.3390/futuretransp5040160>
2. Joshua Kofi Asamoah, Blessing Agyei Kyem, **Nathan David Obeng-Amoako**, Armstrong Aboah, SAAM-ReflectNet: Sign-Aware Attention-Based Multitasking Framework for Integrated Traffic Sign Detection and Retroreflectivity Estimation, *Expert Systems With Applications* (2025), doi: <https://doi.org/10.1016/j.eswa.2025.128003>

Submitted and under review:

- 2026 3. Peter G. Furth, **Nathan David Obeng-Amoako**, Armin Akhavan, Bitá Sadeghinassr, Ray Saeidi, Luis Miranda-Moreno, Christine Théberge-Barrette, Paul Moser, and Nick Ferenchak. Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling. *Journal of Transport Geography*.

RESEARCH REPORTS

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- 2024 1. Eleni Christofa, Chengbo Ai, Peter Furth, Yu-Min Yang, Dewan Tanvir Ahammed, and **Nathan David Obeng-Amoako**. (2024). *Accessible Bus Stop Design in the Presence of Bike Lanes*. Massachusetts Department of Transportation (MassDOT), Office of Transportation Planning. Report No. 24-060. Retrieved from <https://rosap.ntl.bts.gov/view/dot/79327>

JOURNAL REVIEWS

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| 1. Journal of Transportation Engineering, Part A: Systems, ASCE | 2026 – Present |
| 2. Transportation Research Interdisciplinary Perspectives, Elsevier | 2025 – Present |
| 3. Expert Systems with Applications, Elsevier | 2025 – Present |
| 4. Transportation Research Record, SAGE Publications | 2024 – Present |

APPOINTMENTS AND ADVISORY ROLES

Member, Massachusetts Bicycle and Pedestrian Advisory Board, MassDOT	Jan 2026 – Present
- Appointed by Governor Maura T. Healey to advise on state-wide bicycle and pedestrian infrastructure policies and programs - Provide guidance on transportation infrastructure, safety, and accessibility initiatives	

RELEVANT SOFTWARE AND SKILLS

• Python, R, MATLAB	machine learning, deep learning, advanced data analysis
• PTV Vissim/Vistro	traffic flow simulations, traffic signal analysis, traffic impact analysis
• Synchro	traffic flow modeling, traffic signal timing optimization
• ArcGIS	spatial data mapping and analysis for informed decision-making
• Gephi	network data analysis and visualization
• AutoCAD	precision drafting and technical drawings, 2D and 3D
• Bluebeam Revu	creating, editing, and marking up 2D digital blueprints
• HTML/CSS/JavaScript	developing web applications
• Minitab, MS Excel	data visualization, statistical computing, data organization

NOTABLE DISTINCTIONS AND ACHIEVEMENTS

• Grant Recipient, New England African Business Expo, Worcester, MA	May 2025
• Civil & Environmental Engineering Distinguished Fellowship Award, Northeastern Univ.	Aug 2023
• Associated Consultants Ltd. Award for Best Graduating Student, KNUST	Jul 2023
• Provost Award for Best & Excellent Students in College of Engineering, KNUST (x4)	2019 – 2022
• Academic Excellence Award, Ghana Engineering Students' Association, Ghana	Sep 2022
• Global Finalist, Unilever Future Leaders League Business Pitch Competition, U.K.	Jul 2022
• Scholar, Tullow Oil Scholarship Scheme, Ghana	Apr 2019
• Semifinalist, National Science and Maths Quiz (NSMQ), Ghana	Jul 2018

SELECTED CONFERENCE PRESENTATIONS

2026	Obeng-Amoako, N. D., Furth, P., Akhavan, A., Sadeghinassr, B., Saeidi, R., Miranda-Moreno, L., Théberge-Barrette, C., & Moser, P. (2026, August). <i>Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling</i>. Bridging Transportation Researchers (BTR) Conference, Online. - Furth, P., & Obeng-Amoako N. D. (2026, May). <i>Why Johnny and Mary Can't Ride to School: Identifying Barriers to Safe Bicycle Access</i>. Boston Area Research Initiative (BARI) Conference, Boston, MA. - Yang, Y., Ahammed, D. T., Obeng-Amoako, N. D., Christofa, E., Ai, C., & Furth, P. (2026, January). <i>Community-Engaged Informed Floating Bus Stop Guidelines</i> [Poster]. Transportation Research Board Annual Meeting, Washington, DC.
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- 2025 4. **Obeng-Amoako, N. D.**, & Furth, P. (2025, May). *Underlying Traffic Signal Control: The Key to Effective Transit Signal Priority – A Case Study of MBTA Bus Route 39 along South Huntington Avenue, Boston, MA* [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA.
5. Yang, Y., Ahammed, D. T., **Obeng-Amoako, N. D.**, Christofa, E., Ai, C., & Furth, P. (2025, June). *Design Guidelines for Accessible Floating Bus Stops*. International Conference on Transportation and Development (ASCE), Glendale, AZ.
- 2024 6. Yang, Y., Ahammed, D. T., **Obeng-Amoako, N. D.**, Christofa, E., Ai, C., & Furth, P. (2024, April). *Accessible Bus Stop Design in the Presence of Bike Lanes* [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA.
- 2023 7. **Obeng-Amoako, N. D.**, Senaya, A. L., Kassim, B., Owusu-Ansah, F., Damoah, E., Worclachie, S., & Tutu, K. A. (2023, April). *Modification of Bitumen with Waste Materials for Enhanced Aggregate Retention in Surface-Dressed Roads*. 3rd IRF Africa Regional Congress & Exhibition, Accra, Ghana.

LEADERSHIP EXPERIENCE

Secretary, Ghana Transportation Professionals Forum (GTPF)	Feb 2026 – Present
- Maintain membership records and coordinate executive board communications - Support event logistics and engagements with Ghana transportation stakeholders	
Committee Member, Career Mixer Committee, CEE Dep't, Northeastern Univ.	Sep 2025 – Feb 2026
- Organizing networking events connecting students with industry - Coordinating logistics for career dev't workshops and employer engagements	
Executive C'ttee Member, CEE Graduate Student Council, Northeastern Univ.	Feb 2024 – Present
- Brainstorming and implementing strategies to enhance graduate student experience - Serving as a liaison between the departmental chair and the graduate student body	
Department Rep., College of Engineering. PhD Council, Northeastern Univ.	Jun 2024 – Aug 2025
- Meeting with advisor to discuss novel ways of improving student experience - Organizing weekly coffee hours and semesterly happy hours for PhD students	
Communications Director, Young Professionals and Youth Coalition, Ghana	Jan 2021 – Dec 2021
- Social media management to curate consistent communications content - Organization of in-person and virtual engagements with members and partners	
President, Civil Engineering Students' Association, KNUST	Feb 2020 – Sep 2021
- Coordination of fundraising events to secure administrative equipment - Establishment of academic support scheme for underachieving students	

TEACHING EXPERIENCE

Northeastern University, Boston, MA

- Spring 2026 **Graduate Teaching Assistant, Energy Systems: Science, Technology, and Sustainability**
- Provided grading support for coursework on energy technologies and sustainability
 - Gave an invited lecture on Transportation Energy Systems in a Developing World
- Spring 2026 **Graduate Teaching Assistant, Transportation Systems Analysis and Planning**
- Supported students during office hours with questions on design software (Synchro)
 - Evaluated assignments on intersection analysis, signal timing, and street design principles
- Fall 2025 **Graduate Teaching Assistant, Highway Design**
- Provided grading support for homework assignments
 - Assisted students during office hours, addressing questions about design principles
- Fall 2025 **Graduate Teaching Assistant, Transportation Systems Analysis and Planning**
- Evaluated coursework and provided constructive feedback on analytical assignments
 - Supported students' understanding of transportation system planning frameworks
- Fall 2023 **Course Instructor, Recitation for Statics and Solid Mechanics**
- Tutored fundamental concepts of Newtonian physics and solid mechanics
 - Evaluated students' test performance and offered comprehensive feedback

PROFESSIONAL AFFILIATIONS

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| • American Society of Civil Engineers (ASCE) | Feb 2025 – Present |
| • Boston Society of Civil Engineers Section (BSCES) of ASCE | Feb 2025 – Present |
| • New England Intelligent Transportation Society (NEITS) | Sep 2023 – Present |
| • Institute of Transportation Engineers (ITE) | Jan 2024 – Present |

VOLUNTEER EXPERIENCE & COMMUNITY INVOLVEMENT

- **Judge**, Massachusetts Science and Engineering Fair, Worcester (April 2025)
- **Cycling Activist**, Boston Cycling Union, Boston, MA (Feb 2025 – Present)
- **Executive Vice President**, Student Energy Society, KNUST (Jan 2022 – Oct 2022)
- **Community Outreach Coordinator**, Nat'l Society of Black Engineers, KNUST (Jan 2022 – Sep 2022)
- **Project Coordinator**, Engineers Without Borders, KNUST (Aug 2021 - Jul 2022)

BLOG POSTS

1. "Ghana is Moving, but Barely: The Case for a Smarter Public Transportation Strategy." MyJoyOnline, April 2026. <https://www.myjoyonline.com/ghana-is-moving-but-barely-the-case-for-a-smarter-public-transportation-strategy/>

2. “Replacing the Reid Overpass with an At-Grade Intersection (Cambridge, near BU Bridge).” Co-authored with Peter G. Furth. January 2026. <https://peterfurth.sites.northeastern.edu/2026/01/06/replacing-the-reid-overpass-with-an-at-grade-intersection-cambridge-near-bu-bridge/>
3. “PhD Student Reflects on the Sustainable Transportation Dialogue of Civilizations in Holland.” Northeastern University College of Engineering News, August 2024. <https://coe.northeastern.edu/news/a-reflection-on-the-sustainable-transportation-dialogue-of-civilizations-course-in-holland/>
4. “Bus-Friendly Traffic Signals Can Reduce Bus Delay By 90% on South Huntington Avenue.” Co-authored with Peter G. Furth. April 2024. <https://peterfurth.sites.northeastern.edu/2024/06/04/bus-friendly-traffic-signals-can-reduce-bus-delay-by-90-on-south-huntington-avenue/>

REFERENCES

Available upon request